

OSSP PRACTICAL WEEK 3

2520080012

Task 3

```
alurusaranya@LAPTOP-0PA61QEN:/root$ ls
ls: cannot open directory '.': Permission denied
alurusaranya@LAPTOP-0PA61QEN:/root$ cd ~
alurusaranya@LAPTOP-0PA61QEN:~$ pwd
/home/alurusaranya
alurusaranya@LAPTOP-0PA61QEN:~$ nano process_fork.c
alurusaranya@LAPTOP-0PA61QEN:~$ gcc process_fork.c -o process_fork
alurusaranya@LAPTOP-0PA61QEN:~$ ./process_fork
Before fork()

[Parent Process]
PID : 451

PPID : 428
[Child Process]
State: Running
PID : 452
PPID : 451
State: Running
Child exiting
Child terminated
alurusaranya@LAPTOP-0PA61QEN:~$ |
```

Task 3:

```
GNU nano 7.2 process_fork.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {
    pid_t pid;

    printf("Before fork()\n");

    pid = fork();

    if (pid < 0) {
        perror("Fork failed");
        exit(1);
    }
    else if (pid == 0) {
        // Child Process
        printf("\n[Child Process]\n");
        printf("PID : %d\n", getpid());
        printf("PPID : %d\n", getppid());
        printf("State: Running\n");

        // Sleep to allow inspection of waiting/sleeping state
        sleep(10);

        printf("Child exiting\n");
        exit(0);
    }
    else {
        // Parent Process
        printf("\n[Parent Process]\n");
        printf("PID : %d\n", getpid());
        printf("PPID : %d\n", getppid());
        printf("State: Running\n");

        // Wait for child process to finish execution
        wait(NULL);

        printf("Child terminated\n");
        exit(0);
    }

    return 0;
}
```

Task 4

```
alurusaranya@LAPTOP-0PA61QEN:~$ ps -o pid,ppid,state,comm -p <PID>
bash: syntax error near unexpected token `newline'
alurusaranya@LAPTOP-0PA61QEN:~$ ps -l
F S  UID      PID   PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
4 S  1000      428     404  0  80   0  -  1521 do_wai pts/0    00:00:00 bash
0 R  1000      466     428  50  80   0  -  2080 -          pts/0    00:00:00 ps
alurusaranya@LAPTOP-0PA61QEN:~$ top
top - 04:54:49 up 3 min, 2 users, load average: 0.00, 0.00, 0.00
Tasks: 26 total, 1 running, 25 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 99.9 id,  0.0 wa,  0.0 hi,  0.1 si,  0.0 st
MiB Mem : 7712.1 total, 6304.8 free, 491.7 used, 1065.6 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used, 7220.5 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR S %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   21792 13180  9796 S   0.0   0.2   0:00.79 systemd
    2 root        20   0   3180   2200  2072 S   0.0   0.0   0:00.01 init-systemd(Ub
    6 root        20   0   3196   2132  2008 S   0.0   0.0   0:00.00 init
   52 root        19  -1  42256 16332 15256 S   0.0   0.2   0:00.16 systemd-journal
  105 root        20   0  25548  6904  5124 S   0.0   0.1   0:00.16 systemd-udev
  111 systemd+    20   0  21344 13292 11168 S   0.0   0.2   0:00.09 systemd-resolve
  112 systemd+    20   0   91036 7984  7020 S   0.0   0.1   0:00.05 systemd-timesyn
  158 root        20   0   4244   2684  2432 S   0.0   0.0   0:00.01 cron
  159 message+    20   0   9640   5360  4656 S   0.0   0.1   0:00.06 dbus-daemon
  169 root        20   0   18156  8872  7812 S   0.0   0.1   0:00.05 systemd-logind
  171 root        20   0 1755848 13004 11016 S   0.0   0.2   0:00.09 wsl-pro-service
  186 syslog      20   0 222516  5992  4584 S   0.0   0.1   0:00.07 rsyslogd
  193 root        20   0   3124   1972  1832 S   0.0   0.0   0:00.00agetty
  205 root        20   0 107032 23240 13780 S   0.0   0.3   0:00.12 unattended-upgr
  286 root        20   0   3188   1104   980 S   0.0   0.0   0:00.00 SessionLeader
  287 root        20   0   3204   1244  1108 S   0.0   0.0   0:00.03 Relay(288)
  288 root        20   0   6080   5384  3704 S   0.0   0.1   0:00.02 bash
  289 root        20   0   6672   4496  3728 S   0.0   0.1   0:00.01 login
  334 root        20   0 20324 11572  9452 S   0.0   0.1   0:00.08 systemd
  335 root        20   0 21172  3516  1804 S   0.0   0.0   0:00.00 (sd-pam)
  357 root        20   0   6080   5328  3704 S   0.0   0.1   0:00.01 bash
  404 root        20   0   6808   4428  3964 S   0.0   0.1   0:00.00 su
  407 alurusa+    20   0 20344 11528  9388 S   0.0   0.1   0:00.10 systemd
```